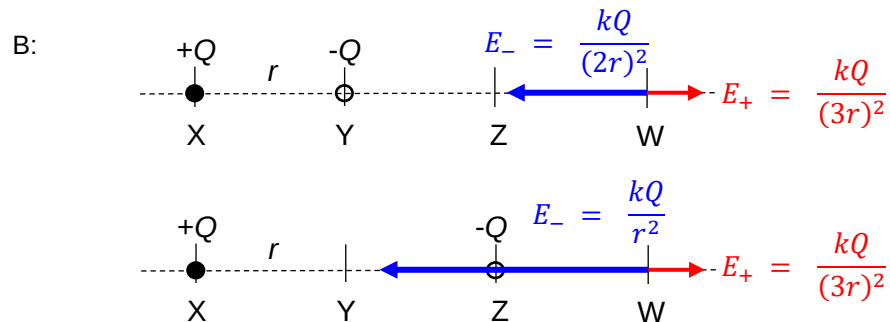


20 C

A: $U = k \frac{Q_1 Q_2}{r_{12}}$. When r increases, U becomes less negative, i.e. the potential energy increases.



C: $V_W = k \frac{Q_1}{r_{1W}} + k \frac{Q_2}{r_{2W}}$. When $-Q$ is moved to Z, which is closer to W, V_W becomes more negative, i.e. the potential at W decreases.

D: $+Q$ and $-Q$ are at the same distance from Y. Based on the formula above, the potential at Y is zero.

21 D

Volume V is constant, express cross sectional area A in terms of V and L : $AL = V \Rightarrow A = V/L$